

Re-annotation Guidelines for Aspect-Level Sentiment Annotation of Agricultural Tweets

The original version of the dataset contained a single sentiment label for each tweet, which was sufficient for sentence-level sentiment analysis but did not provide the granularity required for Aspect-Based Sentiment Analysis (ABSA). To support fine-grained opinion mining, the dataset was enriched through manual aspect-level sentiment annotation, where a sentiment label was assigned to each identified aspect within a tweet. The resulting annotations were subsequently reviewed and validated by an agricultural domain expert from the Institut National Agronomique de Tunisie (INAT) to ensure domain consistency, terminological correctness, and semantic coherence of the annotated aspect-sentiment pairs. This validation process enhanced the reliability of the dataset for agricultural ABSA. The complete annotation guidelines are provided with the dataset to facilitate transparency, reproducibility, and future research.

1. Dataset Overview

The dataset consists of agricultural tweets discussing pesticides, crop diseases, pest control, and farming practices. The tweets cover six agricultural domains:

- Insects
- Fungi
- Weeds
- Bacteria
- Nematodes
- Viruses

Each tweet contains:

- The original tweet text.
- One or more automatically extracted aspect categories.
- A sentiment label to be assigned for each extracted aspect.

The aspect extraction step has already been completed automatically. Therefore, annotators **must not modify, remove, or add aspects**. Their role is exclusively to assign sentiment labels to the provided aspects.

2. Annotation Objective

Each aspect must receive **exactly one** sentiment label selected from the following categories:

- Positive
- Negative
- Neutral

Since a tweet may discuss multiple aspects, multiple sentiment labels may be assigned within the same tweet.

3. Aspect Categories

The annotation considers the following predefined aspect categories.

Effectiveness

Refers to the performance or efficiency of a pesticide, treatment, or agricultural practice.

Examples include:

- effectiveness against pests
- disease control
- treatment success
- product performance

Usage

Refers to the application or utilization of a product.

Examples include:

- application method
- dosage
- spraying
- frequency of use
- ease of application

Pest

Refers to pests, insects, fungi, weeds, bacteria, nematodes, viruses, or crop diseases discussed in the tweet.

Safety

Refers to environmental safety or safety for humans, animals, or beneficial organisms.

Examples include:

- bees
- pollinators
- environment
- farmers
- consumers

Toxicity

Refers to harmful or undesirable effects caused by a pesticide or treatment.

Examples include:

- crop damage
- leaf burn
- poisoning
- destroying

Control Methods

Refers to agricultural strategies used to prevent or control pests or diseases.

Examples include:

- biological and organic control
- integrated pest management
- crop rotation
- cultural practices

4. Sentiment Labels

Positive

Assign "Positive" when the tweet expresses:

- satisfaction,
- approval,
- recommendation,
- successful results,
- beneficial effects, etc.

toward the corresponding aspect.

Example

Tweet:

Neem oil completely eliminated aphids.

Aspect

- Effectiveness

Sentiment

- Positive

Negative

Assign **Negative** when the tweet expresses:

- dissatisfaction,
- criticism,
- failure,
- harmful consequences,
- poor performance, etc.

toward the corresponding aspect.

Example

Tweet:

This fungicide damaged my tomato plants.

Aspect

- Toxicity

Sentiment

- Negative

Neutral

Assign **Neutral** when the tweet:

- provides factual information,
- reports an event,
- asks a question,
- expresses no clear opinion regarding the aspect.

Example

Tweet:

The pesticide was applied yesterday.

Aspect

- Usage

Sentiment

- Neutral

5. Annotation Procedure

For each tweet, follow the steps below.

Step 1

Read the entire tweet carefully to understand its context.

Step 2

Review the automatically extracted aspect list.

Step 3

For each aspect independently, determine the sentiment expressed by the tweet.

Step 4

Assign one sentiment label:

- Positive
- Negative
- Neutral

Step 5

Verify that:

- every aspect has one sentiment label;
- the order of the sentiment labels exactly matches the order of the aspect list.

Step 6

Save the annotation.

6. Annotation Rules

The following rules must be applied consistently.

Rule 1

Do not assign one overall sentiment to the tweet.

Only annotate the provided aspects.

Rule 2

Each aspect is evaluated independently.

Different aspects within the same tweet may receive different sentiment labels.

Rule 3

Do not modify the aspect list.

Specifically:

- do not add aspects;
- do not remove aspects;
- do not rename aspects.

Rule 4

The sentiment labels must follow the same order as the aspect list.

Example

Aspect

effectiveness, toxicity

Sentiment

positive, negative

Rule 5

If no clear opinion is expressed toward an aspect, assign **Neutral**.

7. Difficult Cases

Questions

Questions without an expressed opinion should be annotated as **Neutral**.

Example

Has anyone tried this pesticide?

Future Intentions

Statements describing future actions without expressing an opinion should be annotated as **Neutral**.

Example

I will use this fungicide next week.

Negation

Negation changes the sentiment.

Example

The pesticide did not work.

Effectiveness → Negative

Recommendations

Recommendations generally indicate **Positive** sentiment.

Example

I highly recommend using neem oil.

Effectiveness → Positive

8. Annotation Examples

Example 1

Tweet

This herbicide burned my crops.

Aspect

toxicity

Sentiment

negative

Example 2

Tweet

Applied the pesticide yesterday.

Aspect

usage

Sentiment

neutral

Example 3

Tweet

The pesticide controlled insects but harmed bees.

Aspects

effectiveness, pest, safety

Sentiments

positive, negative, negative